

8.3 Surface Area of Prisms and Pyramids

MPM1D

The surface area of an object is the sum of the areas of its outside surfaces.
(How much material is needed to build an object)

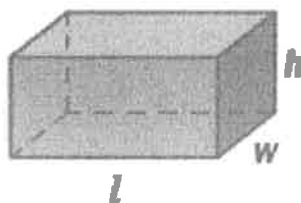
Part 1: Surface Area of Rectangular Prisms

The formula for a rectangular prism is:

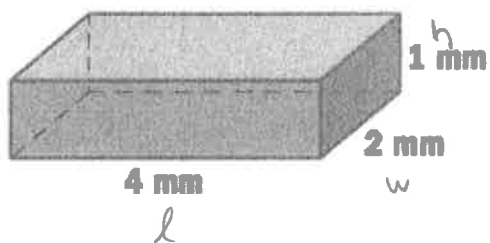
$$S = 2lw + 2wh + 2lh$$

or

$$S = 2(lw + wh + lh)$$

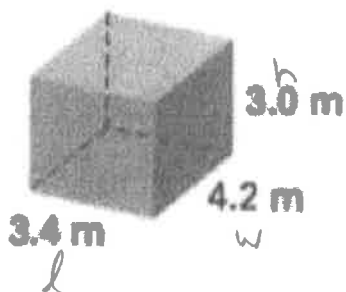


Example 1:



$$\begin{aligned} SA &= 2(4)(2) + 2(2)(1) + 2(4)(1) \\ &= 16 + 4 + 8 \\ &= 28 \text{ mm}^2 \end{aligned}$$

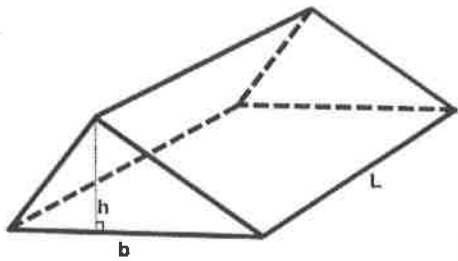
Example 2: What is the total surface area of a room that has the following measurements?



$$\begin{aligned} SA &= 2lw + 2wh + 2lh \\ &= 2(3.4)(4.2) + 2(4.2)(3.0) + 2(3.4)(3) \\ &= 28.56 + 25.2 + 20.4 \\ &= 74.16 \text{ m}^2 \end{aligned}$$

Part 2: Surface Area of Triangular Prisms

How Can We Find the Surface Area of a Triangular Prism?



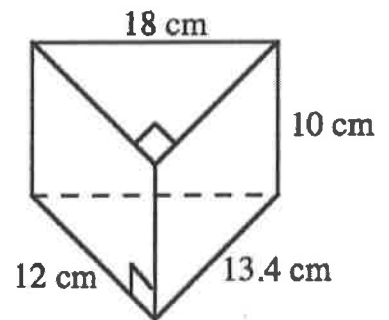
Add the areas of all 5 sides.

Note: The triangular ends are equal in area

$$SA = \text{Area of } \underline{3 \text{ rectangles}} + 2 \text{Area of } \underline{\text{triangle}}$$

Example 3: Find the surface area of the following triangular prism

$$\begin{aligned} \text{Area of triangular base} &= \frac{1}{2}bh \\ &= \frac{1}{2}(12)(13.4) \\ &= 80.4 \text{ cm}^2 \end{aligned}$$



$$\begin{aligned} \text{Area of Rectangle 1} &= lw \\ &= (10)(13.4) \\ &= 134 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area of Rectangle 2} &= lw \\ &= (12)(10) \\ &= 120 \text{ cm}^2 \end{aligned}$$

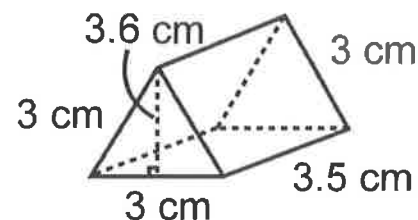
$$\begin{aligned} \text{Area of Rectangle 3} &= lw \\ &= (18)(10) \\ &= 180 \text{ cm}^2 \end{aligned}$$

Total Surface area = Area of rectangles + 2(Area of triangular base)

$$\begin{aligned} &= 134 + 120 + 180 + 2(80.4) \\ &= 434 + 160.8 \\ &= 594.8 \text{ cm}^2 \end{aligned}$$

Example 4: Find the surface area of the following triangular prism

Note: All 3 sides of the triangular base are equal. This means that all three of the rectangular sides will be congruent.



$$\text{Area of triangular base} = \frac{1}{2}bh$$

$$= \frac{1}{2}(3)(3.6)$$

$$= 5.4 \text{ cm}^2$$

$$\text{Area of rectangle} = lw$$

$$= (3.5)(3)$$

$$= 10.5 \text{ cm}^2$$

$$\text{Total Surface Area} = 3(\text{Area of rectangle}) + 2(\text{Area of triangular base})$$

$$= 3(10.5) + 2(5.4)$$

$$= 31.5 + 10.8$$

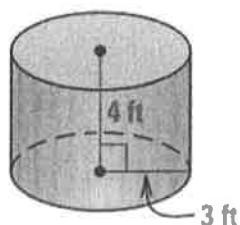
$$= 42.3 \text{ cm}^2$$

Part 3: Surface Area of a Cylinder

$$\text{Surface Area of a Cylinder} = 2(\text{Area of base}) + \text{Area of Lateral Surface}$$

$$\text{Surface Area of a Cylinder} = 2\pi r^2 + 2\pi rh$$

Example 5: Find the surface area of the following cylinder:

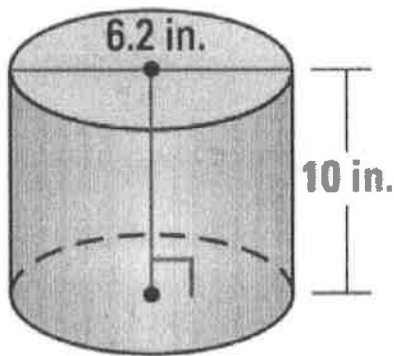


$$SA = 2\pi(3)^2 + 2\pi(3)(4)$$

$$= 56.5 + 75.4$$

$$= 131.9 \text{ ft}^2$$

Example 6: Find the surface area of the following cylinder:



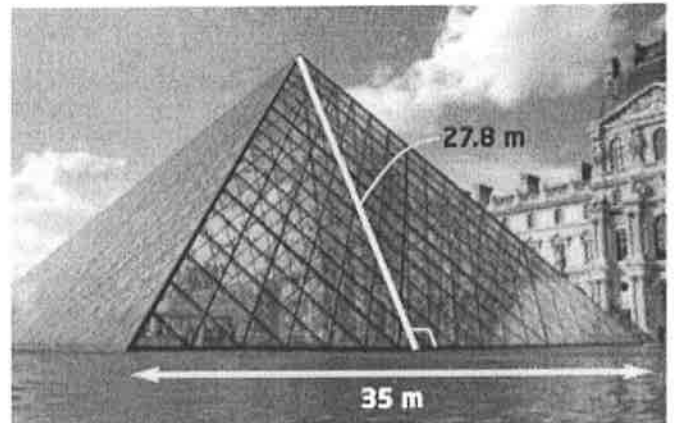
$$\begin{aligned} SA &= 2\pi(3.1)^2 + 2\pi(3.1)(10) \\ &= 60.4 + 194.8 \\ &= 255.2 \text{ in}^2 \end{aligned}$$

Part 4: Surface Area of a Pyramid

Example 7: A modern example of a pyramid can be found at the Louvre in Paris, France. The glass square-based pyramid was built as an entrance to this famous museum. Calculate the surface area of the pyramid including the square base.

$$SA_{\text{pyramid}} = A_{\text{base}} + 4A_{\text{triangle}}$$

Note: This formula only works for a pyramid with a square base



$$\begin{aligned} SA &= (35)^2 + 4\left(\frac{1}{2}\right)(35)(27.8) \\ &= 1225 + 1946 \\ &= 3171 \text{ m}^2 \end{aligned}$$

Homework: Complete Worksheet